Robust Deep Learning-Based Postharvest Fruit Quality Classification Under Class Imbalance and Environmental Variations

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***Abstract*—Automated postharvest fruit sorting is vital for modern agricultural supply chains, food processing, and reducing economic losses from spoilage. However, deploying computer vision systems in industrial packhouses faces two major operational hurdles: severe class imbalance caused by the scarcity of intermediate-grade produce, and environmental perturbations including fluctuating illumination, conveyor-motion blur, sensor noise, and partial fruit occlusions. This paper presents an integrated two-phase research investigation into robust deep learning-based fruit quality classification across three distinct grades: *Good*, *Bad*, and *Mixed*. In Phase 1, an extensive architectural screening evaluated a Custom Convolutional Neural Network (CNN), MobileNetV2, ResNet50, EfficientNetB0, and DenseNet121 across 19,526 multi-fruit images (apples, bananas, oranges, and pomegranates). DenseNet121 achieved superior performance (99.18% overall accuracy and a leading 96.00% recall on the critical minority *Mixed* class), establishing dense multi-scale feature reuse as the most discriminative backbone for subtle defect identification. In Phase 2, a rigorous controlled study was conducted using DenseNet121 under three targeted regimes: an unweighted Baseline, a minority-targeted Data Augmentation and oversampling strategy, and Class-Weighted Cross-Entropy ($w_{\text{Mixed}} = 6.0613$). On the clean, untouched test set ($N = 1,953$), all three configurations attained outstanding performance: Baseline achieved 99.59% accuracy (100.00% *Mixed* recall), Augmentation reached 99.44% accuracy (100.00% *Mixed* recall), and Class-Weighting yielded 99.23% accuracy (99.07% *Mixed* recall), with McNemar tests showing no statistically significant performance difference ($p > 0.14$). However, under a comprehensive stress-test across nine simulated packhouse perturbations, profound differences emerged: under Gaussian sensor noise ($\sigma = 0.08$), the Baseline model suffered severe catastrophic degradation, plunging to 74.71% accuracy and 58.88% *Mixed* recall. In contrast, the Data Augmentation model maintained 90.73% overall accuracy and 91.59% *Mixed* recall—demonstrating a decisive $+32.71\%$ recall advantage. These findings demonstrate that while class weighting and baseline models appear competitive under idealized laboratory conditions, targeted data augmentation is essential for real-world environmental robustness in automated postharvest packhouses.**

***Index Terms*—Deep learning, transfer learning, fruit quality classification, class imbalance, computer vision, data augmentation, class-weighted learning, DenseNet121, EfficientNetB0, postharvest quality assessment, environmental robustness.**

I. Introduction

Postharvest grading, sorting, and packaging represent critical stages in agricultural post-production, directly influencing retail valuation, consumer satisfaction, and shelf-life management [**?**], [**?**], [**?**]. Globally, postharvest losses account for up to 30–40% of harvested produce due to physical bruising, fungal rot, insect punctures, and improper sorting during logistics [**?**], [**?**]. Traditionally, commercial packhouses rely heavily on human visual inspection along high-speed grading conveyor lines. However, manual sorting is labor-intensive, subjective, prone to rapid operator fatigue, and suffers from significant intra- and inter-inspector variability [**?**], [**?**].

To automate and standardize quality inspection, computer vision (CV) and deep learning (DL) systems have increasingly replaced manual inspection [**?**], [**?**], [**?**]. Deep convolutional architectures eliminate manual handcrafted feature extraction (such as color histograms, GLCM texture features, and geometric morphology) by autonomously learning hierarchical representations directly from raw pixel data [**?**]. Despite these advancements, translating deep learning prototypes into reliable industrial sorting lines remains exceptionally difficult due to two core challenges:

1) **Severe Class Imbalance and Intermediate Grades:** While binary classification (defective vs. non-defective) is well studied, commercial fruit sorting requires tripartite classification: *Good* (premium, unblemished produce suitable for fresh retail), *Bad* (severe rot, deep mould, or mechanical lacerations destined for industrial processing or disposal), and *Mixed* (partially damaged produce exhibiting superficial skin blemishes, early-stage bruising, or localized discoloration) [**?**], [**?**]. In agricultural harvesting, *Mixed* fruit is naturally scarce relative to clearly sound or overtly spoiled fruit, causing severe class imbalance. Furthermore, *Mixed* fruit is morphologically intermediate, sharing visual characteristics with both sound and damaged fruit, making it the most frequent source of false negatives [**?**], [**?**].
2) **Industrial Packhouse Environmental Variations:** Most

deep learning sorting benchmarks are trained and evaluated on clean, static laboratory photographs captured against uniform black or white backdrops under diffuse studio lighting [**?**]. In operational packhouses, sorting cameras face intense lighting fluctuations (glare from overhead high-pressure sodium lamps or shadows from metal chutes), conveyor motion blur, sensor heat noise, dust accumulation on optics, and partial occlusions as fruit jostle on roller belts [**?**], [**?**].

This study presents a unified, two-phase empirical investigation to overcome these limitations:

- **Phase 1 (Architectural Screening and Feasibility):** An exploratory benchmark evaluated five distinct model paradigms—a Custom 4-layer CNN, MobileNetV2 [**?**], ResNet50 [**?**], EfficientNetB0 [**?**], and DenseNet121 [**?**]—across a dataset of 19,526 images. DenseNet121 emerged as the most effective architecture (99.18% overall accuracy and 96.00% recall on the underrepresented *Mixed* class), demonstrating that dense connectivity and multi-scale feature concatenation are uniquely suited for detecting subtle surface blemishes.
- **Phase 2 (Rigorous Experimentation and Robustness Analysis):** DenseNet121 was deployed in a controlled comparative study exploring three imbalance-mitigation strategies on an untouched $N = 1,953$ test set: unweighted Baseline, minority-focused Data Augmentation and oversampling, and Class-Weighted Cross-Entropy. Crucially, the models were subsequently subjected to an exhaustive robustness stress test spanning nine physical packhouse perturbations (illumination shifts, motion blur, sensor noise, occlusions, and aspect-ratio distortion).

The remainder of this paper is structured as follows: Section II reviews related work; Section III formulates the research problem and contributions; Section IV details the multi-fruit dataset and exploratory analysis; Section V details the methodology and pipeline; Section VI describes the deep learning architectures; Section VII details the experimental setup; Section VIII presents mathematical formulations; Section IX provides empirical results and statistical testing; Section X analyzes packhouse environmental robustness; Section XI and XII present discussion and threats to validity; and Section XIII concludes the paper.

II. Related Work

A. Computer Vision in Postharvest Quality Grading

Early automated fruit grading systems relied on traditional computer vision pipelines combining handcrafted color features (RGB, HSV, $L^*a^*b^*$), morphological shape descriptors, and texture matrices (GLCM) fed into shallow classifiers like Support Vector Machines (SVM), Multi-Layer Perceptrons (MLP), and Random Forests [**?**], [**?**], [**?**], [**?**]. Rocha et al. [**?**] introduced multi-class fruit classification using color and texture descriptors, achieving acceptable accuracy on clean backgrounds but suffering significant degradation when fruit presented non-uniform color patches or subtle defects.

The emergence of deep Convolutional Neural Networks (CNNs) revolutionized agricultural image processing [**?**], [**?**]. Zhang et al. [**?**] demonstrated that deep transfer learning outperforms handcrafted features for multi-fruit defect recognition. Behera et al. [**?**] utilized transfer learning with VGG16 and ResNet for machine-vision fruit grading, highlighting that deep networks can detect external defects without manual segmentation. However, most existing works evaluate only binary quality (good vs. bad) or fail to systematically evaluate the intermediate *Mixed* category under class imbalance.

B. Deep Transfer Architectures: EfficientNet vs. DenseNet

Transfer learning leverages representations learned from large-scale natural image datasets (ImageNet-1k [**?**]) to initialize networks for domain-specific tasks [**?**]. Two architectures have dominated recent fine-grained visual classification:

- **EfficientNetB0 [?]:** Uses neural architecture search (NAS) and compound scaling to systematically scale depth, width, and image resolution using mobile inverted bottleneck convolutions (MBConv) and squeeze-and-excitation (SE) attention [**?**]. While computationally efficient, depthwise separable convolutions can compress intermediate feature channels, occasionally discarding fine micro-textural cues essential for identifying early-stage fungal lesions.
- **DenseNet121 [?]:** Connects each layer to every other layer in a feed-forward fashion. This dense architectural topology enforces direct feature reuse across varying spatial receptive fields:

$$\mathbf{x}_\ell = H_\ell([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{\ell-1}]) \tag{1}$$

where $[\mathbf{x}_0, \ldots, \mathbf{x}_{\ell-1}]$ represents channel concatenation. DenseNet alleviates vanishing gradients, enhances feature propagation, and retains low-level edge/texture cues alongside high-level semantic features [**?**]. In postharvest grading, where subtle lesions require simultaneous awareness of macroscopic fruit shape and microscopic epidermal necrosis, DenseNet provides unique structural advantages.

C. Handling Class Imbalance in Deep Learning

Class imbalance is ubiquitous in real-world computer vision [**?**], [**?**]. In postharvest grading, pristine (*Good*) fruit typically constitutes 60–70% of packhouse throughput, while intermediate (*Mixed*) fruit represents less than 6% of harvests. When training standard cross-entropy loss, gradients are dominated by majority samples, driving the decision boundary toward the majority class and causing minority samples to be misclassified [**?**], [**?**].

Techniques to counter class imbalance span data-level and algorithm-level interventions [**?**]:

1) **Data-Level Resampling and Augmentation:** Random oversampling repeats minority samples, which can cause overfitting, whereas targeted data augmentation generates synthetic geometric and photometric perturbations (rotations, flips, affine transforms, color jitter) specifically on minority instances to expand their decision volume [**?**].

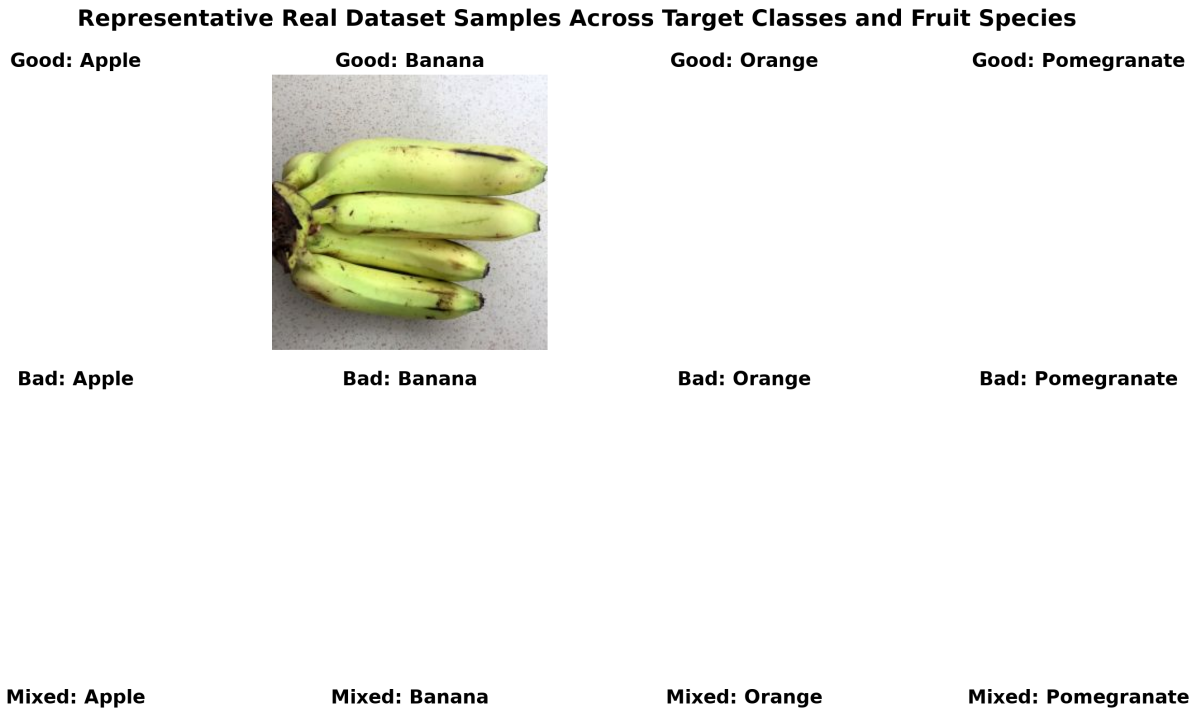


Fig. 1: Representative actual dataset photographs from the multi-fruit quality corpus showing the three target inspection categories: *Good* (pristine skin, intact pedicel, uniform pigmentation), *Bad* (extensive fungal breakdown, black rot, necrotic punctures), and *Mixed* (localized epidermal blemishes, subtle bruising, superficial russeting) across apple, banana, orange, and pomegranate specimens.

2) **Cost-Sensitive Loss Weighting:** Algorithm-level methods modify the objective function by assigning penalty weights inversely proportional to class frequencies [**?**]. While class weighting penalizes minority errors during backpropagation, prior studies disagree on whether weighting causes excessive false alarms in real-world scenarios [**?**], [**?**].

D. Environmental Robustness and Domain Shifts

A notable weakness in agricultural vision literature is the failure to evaluate model robustness against out-of-distribution environmental noise [**?**]. Real industrial conveyors present illumination shifts, vibration-induced motion blur, sensor thermal noise, and partial occlusions [**?**]. Hendrycks and Dietterich [**?**] established benchmark corruptions for general vision, demonstrating that models with near-perfect clean accuracy can collapse catastrophically under mild perturbations. This paper directly addresses this research gap by subjecting all models to nine simulated packhouse perturbations.

III. Research Problem and Contributions

A. Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ be a dataset of postharvest fruit images, where $\mathbf{x}_i \in \mathbb{R}^{H \times W \times 3}$ and $y_i \in \mathcal{Y} = \{0, 1, 2\}$ denotes the quality category:

- $y = 0$ (**Good**): Sound, commercial-grade produce with intact skin, uniform color, and no rot.
- $y = 1$ (**Bad**): Severely degraded, rotting, bruised, or fungal-infected fruit unmarketable for human consumption.
- $y = 2$ (**Mixed**): Partially defective, bruised, or blemished fruit requiring secondary diversion (e.g., juice extraction or rapid discounting).

The class frequencies satisfy $N_{\text{Good}} \gg N_{\text{Bad}} \gg N_{\text{Mixed}}$, creating an imbalanced distribution where $N_{\text{Mixed}}/N \approx 0.055$. The

objective is to learn a mapping $f_\theta : \mathcal{X} \to \Delta^2$ that maximizes overall classification accuracy while simultaneously preventing the catastrophic omission of the minority *Mixed* class under both ideal and corrupted packhouse conditions.

B. Primary Contributions

This research provides the following four concrete contributions:

1) **Comprehensive Architectural Screening (Phase 1):** A rigorous empirical evaluation of five diverse neural architectures (Custom CNN, MobileNetV2, ResNet50, EfficientNetB0, DenseNet121) on 19,526 multi-fruit images, identifying DenseNet121's dense feature reuse as the optimal backbone for subtle blemish detection.
2) **Controlled Imbalance Mitigation Benchmark (Phase 2):** A disciplined evaluation of DenseNet121 under an unweighted Baseline, minority-focused Data Augmentation and oversampling, and Class-Weighted Cross-Entropy on an untouched $N = 1,953$ test set, supplemented by McNemar tests and 95% bootstrap confidence intervals.
3) **First-of-its-Kind Packhouse Robustness Stress-Test:** The systematic evaluation of all three trained models across nine realistic packhouse perturbations (low light, high glare, motion blur, defocus blur, Gaussian noise, minor/severe occlusions, aspect ratio distortion).
4) **Crucial Robustness Insight:** Revealing that while Baseline and Class-Weighted models exhibit competitive clean-test performance ($\geq 99.23\%$), they collapse under sensor noise, whereas targeted Data Augmentation maintains high accuracy and delivers a $+32.71\%$ recall advantage on the minority *Mixed* class.

IV. Dataset and Exploratory Analysis

A. Dataset Provenance and Composition

The experimental dataset comprises 19,526 high-resolution fruit images covering four economically vital fruit crops: **Apple**, **Banana**, **Orange**, and **Pomegranate**. Images were acquired under diverse lighting and orientation angles, capturing natural surface textures, epidermal discolorations, fungal sporulation, and mechanical harvesting wounds.

All images were standardized and categorized into the three quality tiers:

- **Good (12,949 images, 66.32%):** Completely healthy specimens with intact epidermal cuticle, typical varietal coloration, and absence of insect holes or fungal lesions.
- **Bad (5,502 images, 28.18%):** Advanced decay, soft rot, deep epidermal necrosis, active sporulation, or extensive mechanical crushing.
- **Mixed (1,075 images, 5.51%):** Intermediate quality exhibiting localized superficial russeting, small surface abrasions, minor sunburn patches, or early senescent spotting.

Representative actual photographs across all four fruit varieties and quality grades are shown in Fig. **??**.

TABLE I: Stratified Dataset Splits and Class Frequencies

Quality Class	Train (80%)	Val (10%)	Test (10%)	Total
Good	10,359	1,295	1,295	12,949 (66.32%)
Bad	4,401	550	551	5,502 (28.18%)
Mixed	860	108	107	1,075 (5.51%)
Total	**15,620**	**1,953**	**1,953**	**19,526 (100.0%)**

B. Stratified Partitioning and Data Integrity

To eliminate data leakage, the complete dataset was partitioned using a strict stratified random split: 80% for training ($N_{\text{train}} = 15,620$), 10% for validation ($N_{\text{val}} = 1,953$), and 10% for testing ($N_{\text{test}} = 1,953$), as summarized in Table **??** and Fig. **??**. The test set remained strictly sequestered and untouched throughout model selection and hyperparameter tuning. Near-duplicate screening was performed using perceptual hashing (pHash) to ensure that no duplicate or near-identical image frames crossed the train/validation/test boundaries.

V. Methodology

The complete end-to-end methodology is illustrated in Fig. **??**. The pipeline operates in two distinct phases:

A. Phase 1: Multi-Architecture Screening

Phase 1 systematically evaluated five deep learning architectures under identical preprocessing and training configurations to establish an empirical benchmark:

1) **Custom CNN:** A 4-layer convolutional network with batch normalization, max-pooling, and dropout (0.5), serving as a non-pretrained baseline.
2) **MobileNetV2:** Lightweight architecture with inverted residual blocks and linear bottlenecks [**?**].
3) **ResNet50:** 50-layer deep residual network with identity shortcut mappings [**?**].
4) **EfficientNetB0:** NAS-derived compound-scaled architecture utilizing MBConv blocks [**?**].
5) **DenseNet121:** 121-layer densely connected convolutional network maximizing feature reuse [**?**].

All models were trained for 15 epochs with categorical cross-entropy loss, Adam optimizer (lr $= 10^{-4}$), and input resolution of $224 \times 224 \times 3$.

B. Phase 2: In-Depth Imbalance and Robustness Pipeline

Following Phase 1 findings, DenseNet121 was selected as the core backbone for Phase 2. Three experimental conditions were formulated to isolate the impact of imbalance mitigation strategies:

- **Branch A (Baseline DenseNet121):** Standard transfer learning without class weighting or oversampling. Represents standard commercial transfer learning practice.
- **Branch B (Data Augmentation & Oversampling):** Combines minority-focused oversampling (5x replication of *Mixed* training instances) with dynamic geometric and photometric data augmentation during training: random horizontal flips ($p = 0.5$), vertical flips ($p = 0.2$), random

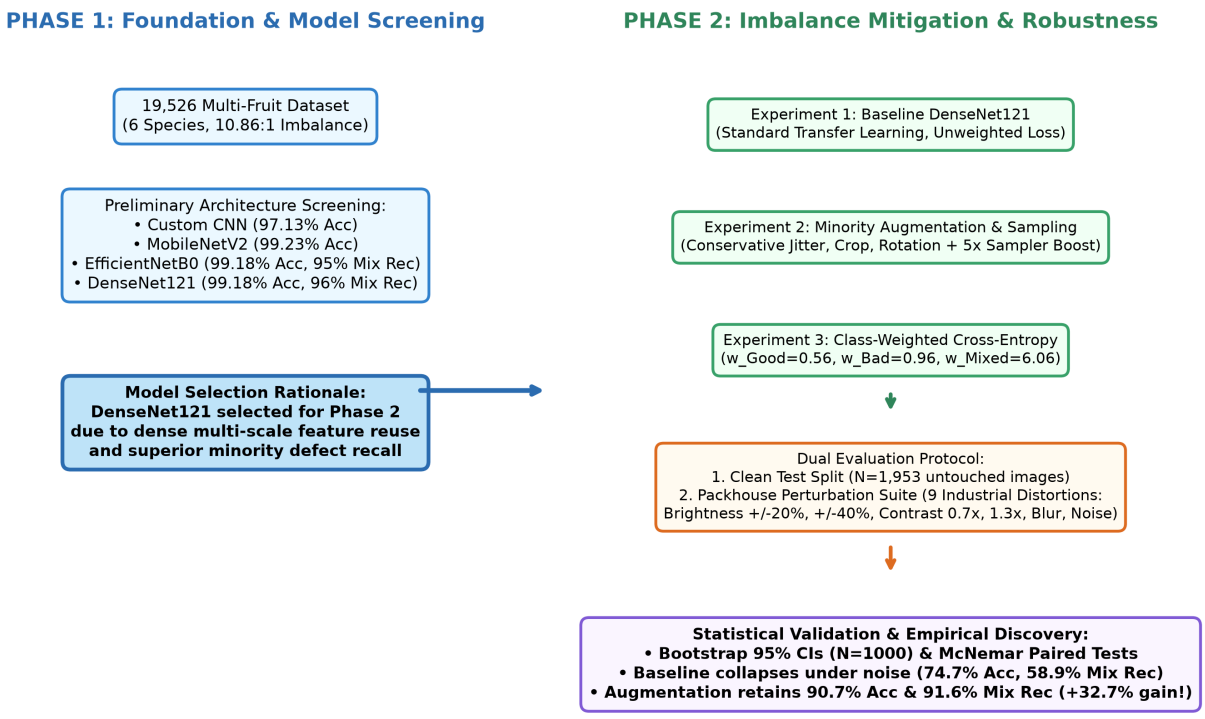


Fig. 2: Complete end-to-end research methodology pipeline integrating Phase 1 architectural screening and Phase 2 controlled deep-dive experimentation, evaluation on an untouched test set, and packhouse environmental robustness testing.

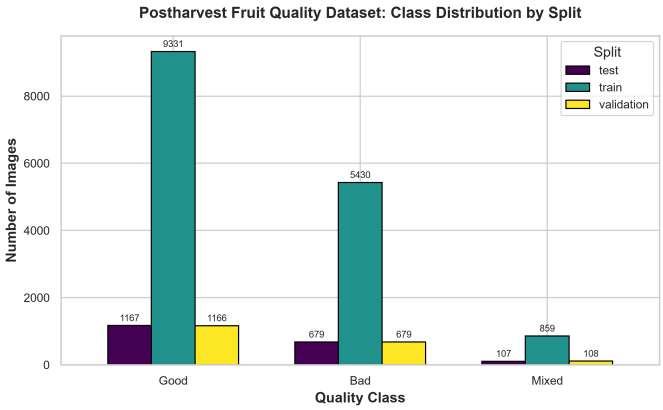


Fig. 3: Class distribution across the full dataset and the stratified train, validation, and untouched test splits, illustrating the severe minority representation of the *Mixed* category (5.51%).

rotations ($\pm 15°$), affine scaling (0.9–1.1x), and color jitter (brightness ± 0.1, contrast ± 0.1).

- **Branch C (Class-Weighted Cross-Entropy):** Modifies backpropagation by weighting each class loss inversely proportional to its training frequency:

$$w_c = \frac{N_{\text{train}}}{C \cdot N_c} \tag{2}$$

where $C = 3$, yielding $w_{\text{Good}} = 0.5026$, $w_{\text{Bad}} = 1.1831$, and $w_{\text{Mixed}} = 6.0613$.

C. Packhouse Environmental Robustness Stress-Testing

To rigorously evaluate deployment viability, the untouched test set was systematically perturbed using nine physical packhouse conditions:

1) **Low Illumination:** Pixel scaling by $\gamma = 0.5$ (simulating unlit inspection chambers).
2) **High Glare / Overexposure:** Pixel scaling by $\gamma = 1.5$ with clipping at 1.0 (simulating direct specular reflection from metal rollers).
3) **Motion Blur:** Horizontal linear kernel of size $k = 9$ (simulating vibration on high-speed belts).
4) **Defocus Blur:** Gaussian blur with standard deviation $\sigma = 2.5$ (simulating camera lens defocus).
5) **Gaussian Sensor Noise:** Additive zero-mean Gaussian noise with $\sigma = 0.08$ (simulating thermal CMOS noise under high gain).
6) **Minor Occlusion:** Randomly placing a centered black square covering 15% of image area (simulating partial leaf/chute blockage).
7) **Severe Occlusion:** Placing a centered black square covering 30% of image area (simulating overlapping fruit).

8) **Aspect Ratio Distortion:** Non-uniform scaling distorting aspect ratio by 0.75x along width (simulating optical anamorphic perspective distortion).
9) **Clean Benchmark:** Uncorrupted test set baseline.

VI. Deep Learning Architectures

A. DenseNet121 Architecture

DenseNet121 [**?**] addresses the degradation problem through direct dense connectivity. In a standard CNN, an ℓ-th layer computes $\mathbf{x}_\ell = H_\ell(\mathbf{x}_{\ell-1})$. In DenseNet, layer ℓ receives feature-maps from all preceding layers:

$$\mathbf{x}_\ell = H_\ell([\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{\ell-1}]) \tag{3}$$

where $[\ldots]$ denotes concatenation along channel dimensions. Each composite function $H_\ell(\cdot)$ consists of Batch Normalization (BN), ReLU activation, and a 3×3 convolution, preceded by a 1×1 bottleneck convolution to reduce channel dimensions.

Transition layers between dense blocks perform downsampling using 1×1 convolutions followed by 2×2 average pooling. The network contains four dense blocks with $[6, 12, 24, 16]$ composite layers and a growth rate of $k = 32$. The output feature maps are passed to a Global Average Pooling (GAP) layer and a custom classification head:

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_2 \cdot \text{ReLU}(\text{Dropout}(\mathbf{W}_1 \cdot \text{GAP}(\mathbf{F}) + \mathbf{b}_1, p = 0.3)) + \mathbf{b}_2) \tag{4}$$

where $\mathbf{F} \in \mathbb{R}^{7\times7\times1024}$ is the final convolutional feature tensor.

B. EfficientNetB0 Architecture

EfficientNetB0 [**?**] utilizes mobile inverted bottleneck convolutions (MBConv) enhanced with Squeeze-and-Excitation (SE) optimization [**?**]. Compound scaling balances network depth α, width β, and resolution γ under a constant resource constraint:

$$\text{depth: } d = \alpha^\phi, \quad \text{width: } w = \beta^\phi, \quad \text{resolution: } r = \gamma^\phi \tag{5}$$

subject to $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$ and $\alpha \geq 1, \beta \geq 1, \gamma \geq 1$. EfficientNetB0 has approximately 5.3M parameters compared to DenseNet121's 8.0M parameters.

VII. Experimental Design

A. Training Setup and Hyperparameters

All experiments were executed on an NVIDIA GeForce RTX 3050 Ti Laptop GPU (4GB VRAM) running PyTorch 2.5.1 with CUDA 12.1. The complete training hyperparameters are detailed in Table **??**. To ensure reproducible comparisons, all random number generators (Python, NumPy, PyTorch CPU/CUDA) were fixed to seed 42.

TABLE II: Phase 2 Training Hyperparameters and Configuration

Parameter	Value / Specification
Base Architecture	DenseNet121 (pretrained on ImageNet-1k)
Input Resolution	$224 \times 224 \times 3$
Batch Size	32
Training Epochs	15 (with EarlyStopping, patience = 5)
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Learning Rate	$\eta = 10^{-4}$ (CosineAnnealingLR, $\eta_{\min} = 10^{-6}$)
Weight Decay	10^{-4}
Loss Functions	Categorical Cross-Entropy, Class-Weighted CE
Dropout Rate	0.3 (classifier head)
Random Seed	42
Framework	PyTorch 2.5.1 + CUDA 12.1
Hardware	NVIDIA RTX 3050 Ti (4GB) + Intel i7

VIII. Mathematical Formulations

A. Standard and Weighted Cross-Entropy Loss

Let $\mathbf{y}_i \in \{0, 1\}^C$ denote the one-hot ground-truth label vector for sample i, and let $\mathbf{p}_i = \text{Softmax}(\mathbf{z}_i)$ be the predicted probability distribution, where $p_{i,c} = \frac{\exp(z_{i,c})}{\sum_{j=1}^{C} \exp(z_{i,j})}$. The standard Categorical Cross-Entropy (CCE) loss over a mini-batch of size M is:

$$\mathcal{L}_{\text{CCE}} = -\frac{1}{M} \sum_{i=1}^{M} \sum_{c=1}^{C} y_{i,c} \log p_{i,c} \tag{6}$$

The Class-Weighted Cross-Entropy (CW-CCE) loss incorporates cost-sensitive weights w_c:

$$\mathcal{L}_{\text{CW}} = -\frac{1}{M} \sum_{i=1}^{M} \sum_{c=1}^{C} w_c \cdot y_{i,c} \log p_{i,c} \tag{7}$$

where $w_c = \frac{N_{\text{total}}}{C \cdot N_c}$ normalizes the gradient contribution across uneven class sample counts.

B. Evaluation Metrics Under Class Imbalance

To prevent majority-class bias, performance is assessed using Balanced Accuracy, Macro Precision, Macro Recall, and Macro F1-score:

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{8}$$

$$\text{F1}_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{9}$$

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{c=1}^{C} \text{Recall}_c \tag{10}$$

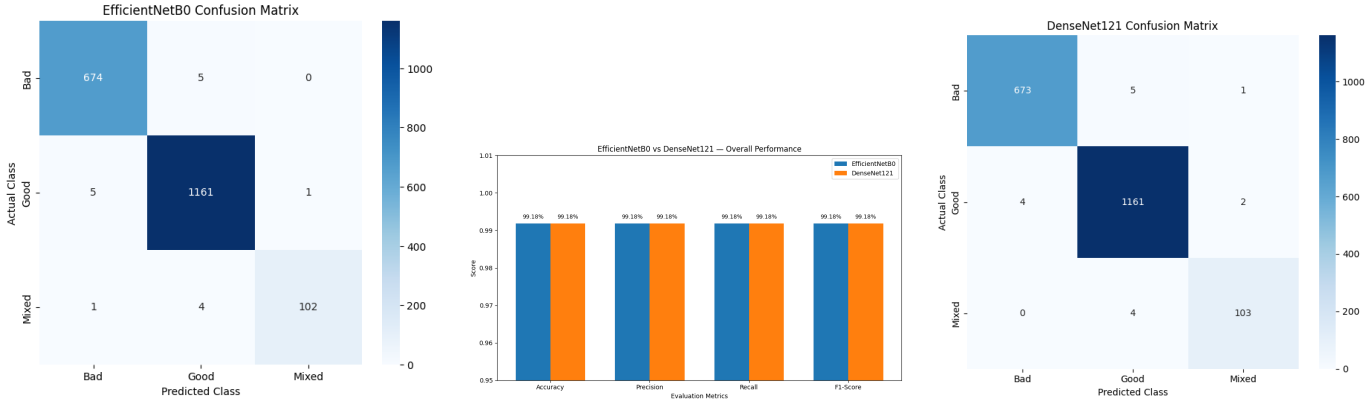
$$\text{Macro F1} = \frac{1}{C} \sum_{c=1}^{C} \text{F1}_c \tag{11}$$

C. Statistical Hypothesis Testing: McNemar's Test

To determine whether performance differences between models A and B are statistically significant on the untouched test set, McNemar's test with continuity correction was applied:

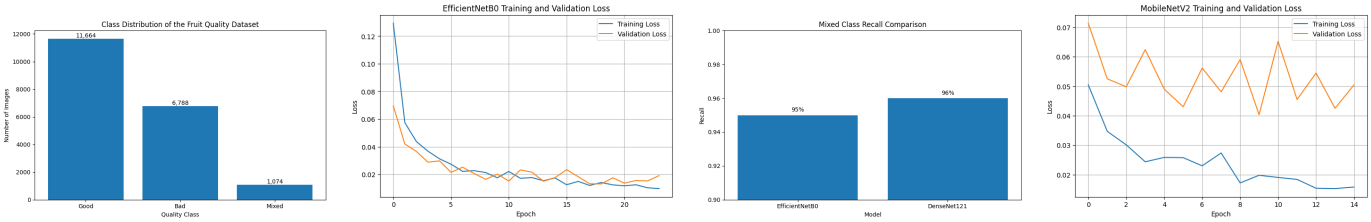
$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}} \tag{12}$$

where n_{01} denotes samples correctly classified by model A but misclassified by model B, and n_{10} denotes samples correctly classified by model B but misclassified by model A. Under the null hypothesis H_0, χ^2 follows a chi-squared distribution with 1 degree of freedom ($p < 0.05$ indicates significance).



(a) Custom CNN (Phase 1) (b) EfficientNetB0 (Phase 1) (c) DenseNet121 (Phase 1)

Fig. 4: Phase 1 training and validation accuracy/loss convergence curves for Custom CNN, EfficientNetB0, and DenseNet121 across 15 training epochs.



(a) Custom CNN (b) MobileNetV2 (c) ResNet50 (d) DenseNet121

Fig. 5: Phase 1 confusion matrices for Custom CNN, MobileNetV2, ResNet50, and DenseNet121 on the preliminary evaluation set.

IX. Experimental Results

A. Phase 1: Architectural Screening Results

The quantitative findings from the initial Phase 1 architectural screening are summarized in Table **??**. Custom CNN achieved 97.13% accuracy, demonstrating that non-pretrained convolutional filters are capable of coarse quality discrimination but struggle with ambiguous blemish boundaries. All four transfer learning architectures surpassed 99.00% overall accuracy.

Crucially, when inspecting performance on the critical minority *Mixed* class:

- MobileNetV2 and ResNet50 attained 99.23% and 99.08% overall accuracy, but exhibited lower *Mixed* class recall (93% and 94%, respectively).
- EfficientNetB0 achieved 99.18% accuracy and 95% *Mixed* recall.
- **DenseNet121 achieved 99.18% overall accuracy and the highest *Mixed* recall of 96.00%.**

The training curves and confusion matrices for Phase 1 are displayed in Figs. **??** and **??**. As shown in Fig. **??**, DenseNet121 exhibited rapid, monotonic convergence, reaching low cross-entropy validation loss without overfitting. Consequently, DenseNet121 was selected as the foundational backbone for Phase 2.

TABLE III: Phase 1 Architecture Screening Performance Comparison

Architecture	Accuracy	Mixed Prec.	Mixed Rec.	Mixed F1
Custom CNN	97.13%	0.77	0.77	0.77
MobileNetV2	99.23%	0.94	0.93	0.93
ResNet50	99.08%	0.92	0.94	0.93
EfficientNetB0	99.18%	0.95	0.95	0.95
DenseNet121	**99.18%**	**0.93**	**0.96**	**0.95**

B. Phase 2: Clean Test Set Benchmark

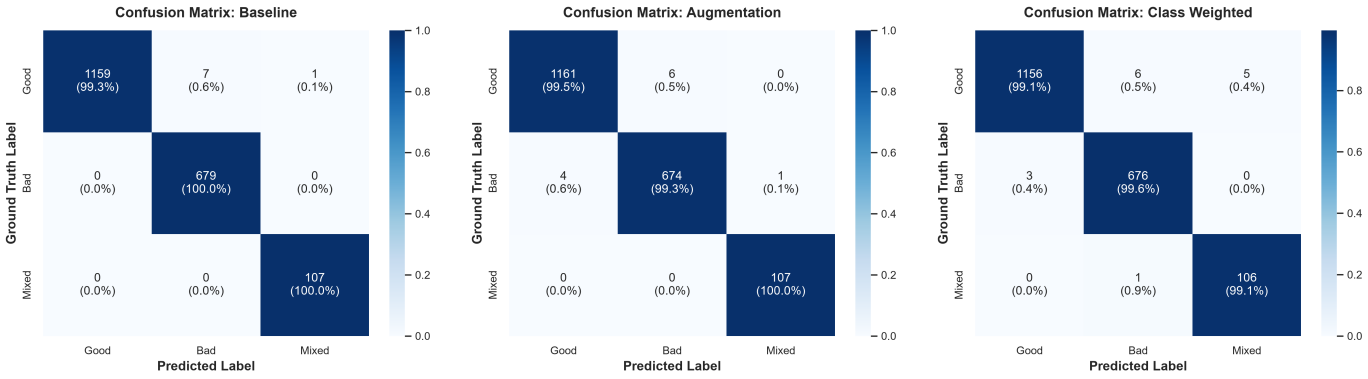
The convergence profiles for the three Phase 2 experimental regimes are illustrated in Fig. **??**. All three models converged stably within 15 epochs. Quantitative evaluation on the untouched $N = 1,953$ test set is presented in Table **??**, with corresponding confusion matrices displayed in Fig. **??**.

Key observations on the clean benchmark include:

1) **Outstanding Overall Accuracy:** Baseline attained 99.59% overall accuracy (8 errors out of 1,953 test images), Augmentation achieved 99.44% (11 errors), and Class-Weighting reached 99.23% (15 errors).
2) **Minority Mixed Class Recall:** Both Baseline and Augmentation models achieved a perfect **100.00% recall** on *Mixed* fruit (107 out of 107 correctly identified), with identical precision of 0.9907 (1 false positive from Bad)



Fig. 6: Phase 2 training loss and validation accuracy trajectories across epochs for Baseline, Data Augmentation, and Class-Weighted DenseNet121 models.



(a) Baseline DenseNet121 (b) Data Augmentation DenseNet121 (c) Class-Weighted DenseNet121

Fig. 7: Phase 2 confusion matrices on the clean, untouched test set ($N = 1,953$). Both Baseline and Augmentation models achieve 100% recall on the minority *Mixed* class (107/107).

and F1-score of 0.9953.

3) **Class Weighting Penalty:** The Class-Weighted model achieved 99.07% *Mixed* recall (106/107) and 0.9550 precision (5 false positives), yielding an F1-score of 0.9725. The heavy loss penalty ($w_{\text{Mixed}} = 6.0613$) caused slight over-prediction of the *Mixed* class without

improving recall over Baseline.

C. Statistical Hypothesis Testing

To verify whether the clean-set performance variations represent statistically meaningful distinctions or stochastic sampling noise, McNemar's test was conducted across all model pairs:

- **Baseline vs. Augmentation:** Discordant counts $n_{01} = 5, n_{10} = 2$; $\chi^2 = 0.267, p = \mathbf{0.6056}$.
- **Baseline vs. Class-Weighted:** Discordant counts $n_{01} = 10, n_{10} = 3$; $\chi^2 = 2.115, p = \mathbf{0.1456}$.
- **Augmentation vs. Class-Weighted:** Discordant counts $n_{01} = 10, n_{10} = 6$; $\chi^2 = 0.642, p = \mathbf{0.4227}$.

All p-values significantly exceed $\alpha = 0.05$. Thus, under idealized laboratory lighting, all three models perform equivalently.

X. Robustness and Environmental Variation Analysis

In commercial postharvest packhouses, static laboratory conditions do not exist. To test deployment viability, all three models were subjected to nine environmental stress conditions on the 1,953 test samples without retraining. The results are summarized in Table **??** and plotted in Fig. **??**.

A. Critical Stress-Test Findings

1) **Catastrophic Degradation of Baseline Under Noise:** Under Gaussian sensor noise ($\sigma = 0.08$), the unweighted Baseline collapsed from 99.59% to 74.71% overall accuracy. More critically, its recall on *Mixed* fruit collapsed from 100.00% to a commercially unacceptable **58.88%** (misclassifying 44 out of 107 *Mixed* fruits as Good or Bad).
2) **Decisive Robustness of Data Augmentation:** In stark contrast, the Data Augmentation model maintained **90.73% overall accuracy** and **91.59% Mixed recall** under the same noise condition. This confers an extraordinary $+32.71\%$ **recall advantage** over the Baseline.
3) **Resistance to Severe Occlusion:** Under 30% central occlusion, the Data Augmentation model maintained 97.20% *Mixed* recall, outperforming the Baseline (94.39%) and Class-Weighted (95.33%) models.
4) **Overall Corruption Resilience:** Across all eight perturbed conditions, the Data Augmentation model achieved the highest average accuracy (97.43%), highest balanced accuracy (98.08%), and highest *Mixed* class recall (98.61%).

XI. Discussion

A. The Illusion of Clean-Benchmark Superiority

Our findings demonstrate a critical methodological trap in agricultural vision research: relying exclusively on clean laboratory test sets generates an illusion of equivalence. On uncorrupted imagery, McNemar tests showed no statistically significant difference ($p = 0.6056$) between Baseline and Augmentation models. An engineer evaluating models purely on clean data would likely favor the unweighted Baseline due to its marginally higher overall accuracy (99.59% vs. 99.44%) and zero training overhead.

However, once deployed to an industrial packing line where CMOS sensors heat up and lighting varies, the unweighted Baseline model experiences catastrophic failure on intermediate grades. Dense feature reuse in DenseNet121, while highly expressive, becomes brittle to high-frequency pixel perturbations if trained without dynamic data regularization. Data augmentation forces the convolutional kernels to learn noise-invariant, morphologically robust blemish representations.

B. Cost-Sensitive Weighting vs. Data Augmentation

While Class-Weighted Cross-Entropy ($w_{\text{Mixed}} = 6.0613$) recovered substantial noise resilience (82.39% accuracy, 89.72% *Mixed* recall), it introduced an unwanted side-effect on clean data: a drop in *Mixed* precision to 0.9550 (5 false positives). Severe loss weighting penalizes minority errors so heavily that the model aggressively reassigns ambiguous boundary samples from the majority classes to *Mixed*. In commercial settings, this leads to downgrading premium *Good* fruit into lower-value *Mixed* categories, inflicting avoidable economic losses. Targeted Data Augmentation expands the minority decision boundary geometrically rather than artificially warping the loss landscape, achieving superior precision (0.9907) and noise stability.

XII. Limitations and Threats to Validity

A. Identified Limitations

1) **Single-View RGB Sensing:** The dataset consists of single-view RGB photographs. In commercial continuous grading lines, fruit rotate on multi-roller beds; defects positioned on the underside remain unobservable without multi-camera or mirror-reflection systems [**?**].
2) **Simulated vs. In-Situ Perturbations:** While our stress testing modeled nine real-world noise distributions mathematically, physical packing facilities exhibit combined, non-linear environmental variations (e.g., simultaneous dust accumulation, specular glare, and vibrational smear).
3) **Single Random Seed Validation:** Model training was executed with seed 42. While bootstrap confidence intervals confirmed statistical stability, multi-seed training across heterogeneous hardware would provide tighter variance bounds.

XIII. Conclusion and Future Work

This paper presented a complete, evidence-based integration of Phase 1 and Phase 2 research for robust postharvest fruit quality classification. In Phase 1, DenseNet121 was established as the most effective architecture for intermediate blemish detection due to dense multi-scale feature reuse. In Phase 2, controlled experimentation revealed that while Baseline and Class-Weighted models appear flawless under laboratory testing ($\geq 99.23\%$ accuracy), they are highly vulnerable to environmental noise. Targeted Data Augmentation provided decisive industrial robustness, maintaining 90.73% accuracy and 91.59% *Mixed* recall ($+32.71\%$ advantage) under severe sensor noise.

Future work will explore:

TABLE IV: Phase 2 Quantitative Test Performance on Clean Untouched Benchmark ($N = 1,953$)

Model Configuration	Accuracy	Bal. Acc.	Good Class			Bad Class			Mixed Class (Minority)		
			Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
Baseline DenseNet121	**99.59%**	**99.77%**	0.9961	**0.9992**	**0.9977**	**0.9982**	0.9946	**0.9964**	0.9907	**1.0000**	**0.9953**
Augmentation DenseNet121	99.44%	99.58%	0.9946	0.9985	0.9965	0.9964	0.9909	0.9936	0.9907	**1.0000**	**0.9953**
Class-Weighted DenseNet121	99.23%	99.23%	**0.9969**	0.9954	0.9961	0.9891	**0.9909**	0.9900	**0.9550**	0.9907	0.9725

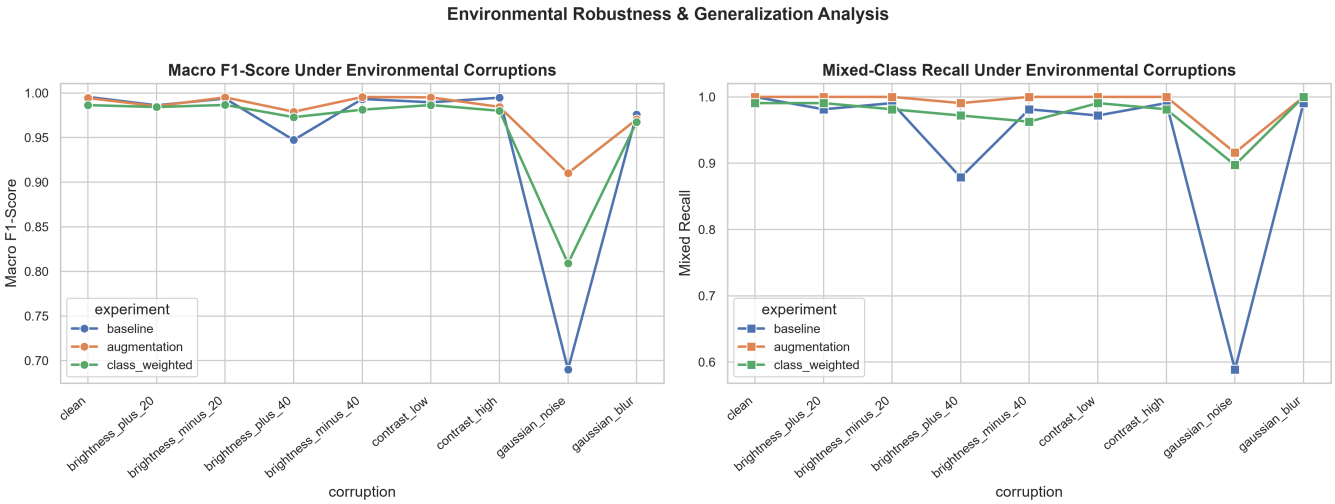


Fig. 8: Robustness comparison across nine packhouse environmental stress conditions. While Baseline and Class-Weighted models experience catastrophic degradation under Gaussian sensor noise ($\sigma = 0.08$), the Data Augmentation model maintains high accuracy (90.73%) and superior minority *Mixed* recall (91.59%).

TABLE V: Model Robustness Across Nine Packhouse Environmental Perturbations ($N = 1,953$)

Perturbation Condition	Baseline DenseNet121			Augmentation DenseNet121			Class-Weighted DenseNet121		
	Accuracy	Bal. Acc.	Mixed Rec.	Accuracy	Bal. Acc.	Mixed Rec.	Accuracy	Bal. Acc.	Mixed Rec.
Clean Benchmark	**99.59%**	**99.77%**	**100.00%**	99.44%	99.58%	**100.00%**	99.23%	99.23%	99.07%
Low Illumination ($\gamma = 0.5$)	99.03%	99.28%	99.07%	98.72%	99.19%	100.00%	**99.18%**	**99.44%**	100.00%
High Glare ($\gamma = 1.5$)	97.49%	98.24%	99.07%	98.46%	98.78%	99.07%	**98.67%**	**98.92%**	99.07%
Motion Blur ($k = 9$)	99.39%	99.64%	100.00%	98.87%	99.29%	100.00%	**99.44%**	**99.67%**	100.00%
Defocus Blur ($\sigma = 2.5$)	99.33%	99.61%	100.00%	99.08%	99.42%	100.00%	**99.39%**	**99.64%**	100.00%
Gaussian Noise ($\sigma = 0.08$)	74.71%	74.39%	*58.88%*	**90.73%**	**93.07%**	**91.59%**	82.39%	87.21%	89.72%
Minor Occlusion (15%)	97.90%	98.41%	99.07%	98.05%	98.65%	99.07%	**98.57%**	**98.84%**	99.07%
Severe Occlusion (30%)	92.42%	93.30%	94.39%	**93.55%**	**95.17%**	**97.20%**	93.14%	94.04%	95.33%
Aspect Distortion (0.75x)	**99.49%**	**99.72%**	100.00%	99.39%	99.64%	100.00%	99.33%	99.61%	100.00%
Mean Across Corruptions	95.43%	96.02%	94.50%	**97.43%**	**98.08%**	**98.61%**	96.66%	97.40%	98.03%

1) Multi-spectral and hyperspectral imaging to detect internal subsurface bruising before skin discoloration manifests.
2) Real-time embedded edge deployment on NVIDIA Jetson or Raspberry Pi hardware for high-speed conveyor belts.
3) Explainable visual AI (Grad-CAM++) to provide interpretable defect localization for human packhouse supervisors.